Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Half Past Paper – Week 26

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
2a	Substituting into real-life formulae	2	U144
2b	Changing the subjects of formulae	2	U556
4a	Solving inverse proportion word problems	2	U357
4b	Solving inverse proportion word problems	1	U357
6	Solving simultaneous equations using elimination or substitution	3	U760,U757
8a	Using the exact values of trigonometric ratios (Higher)	1	U319
8b	Finding unknown sides in right-angled triangles	2	U283
10a	Simplifying algebraic fractions by factorising into one bracket	1	U437
10b	Factorising the difference of two squares	2	U963
12	Mixed problems: Circle theorems	3	U808
14a	Indices of the form a/b	2	U772
14b	Index rules with positive and negative indices, Indices of the form 1/a	2	U235,U694, U985
16	Converting recurring decimals to fractions	3	U689
18a	Transforming graphs	1	U455
18b	Transforming graphs	1	U455
20	Rationalising denominators containing two terms	3	U281
22	Tree diagrams, Algebraic fractions, Factorise to solve quadratics	6	U729,U685, U457,U228
	Total	37	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

2 $v^2 = u^2 + 2as$

$u = 12 \quad a = -3 \quad s = 18$

(*a*) Work out a value of v.

.....

(2)

(*b*) Make s the subject of $v^2 = u^2 + 2as$

.....

(2)

(Total for Question 2 is 4 marks)

4 It would take 120 minutes to fill a swimming pool using water from 5 taps.

(*a*) How many minutes will it take to fill the pool if only 3 of the taps are used?

..... minutes

(2)

(*b*) State one assumption you made in working out your answer to part (a).

.....

.....

(1)

(Total for Question 4 is 3 marks)

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6 Solve the simultaneous equations

$$5x + y = 21$$
$$x - 3y = 9$$

$x =$ .....

$y =$ .....

(Total for Question 6 is 3 marks)

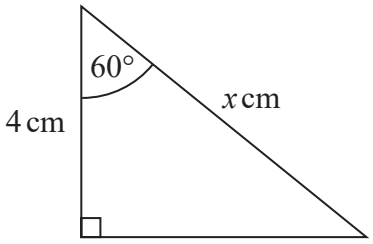
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8 (*a*) Write down the exact value of tan 45°

.....

(1)

Here is a right-angled triangle.



cos 60° = 0.5

(*b*) Work out the value of *x*.

.....

(2)

(Total for Question 8 is 3 marks)

10 (*a*) Simplify $\dfrac{x-1}{5(x-1)^2}$

.....

(1)

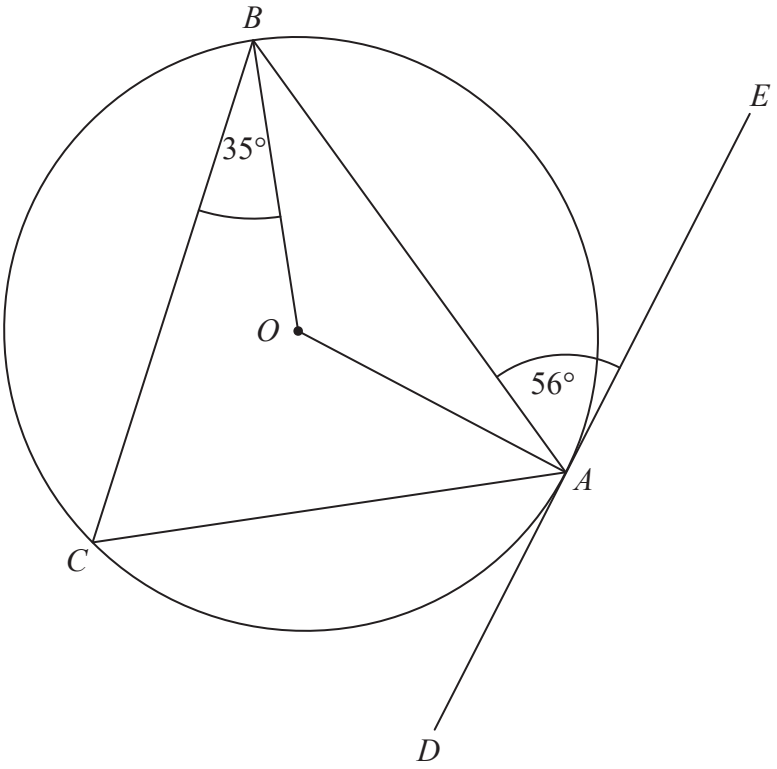
(*b*) Factorise fully $50 - 2y^2$

.....

(2)

(Total for Question 10 is 3 marks)

12



A, *B* and *C* are points on the circumference of a circle, centre *O*.
DAE is the tangent to the circle at *A*.

Angle *BAE* = 56°
Angle *CBO* = 35°

Work out the size of angle *CAO*.
You must show all your working.

.....°

(Total for Question 12 is 3 marks)

14 (*a*) Work out the value of $\left(\frac{16}{81}\right)^{\frac{3}{4}}$

.....

(2)

$3^a = \frac{1}{9}$ $3^b = 9\sqrt{3}$ $3^c = \frac{1}{\sqrt{3}}$

(*b*) Work out the value of $a + b + c$

.....

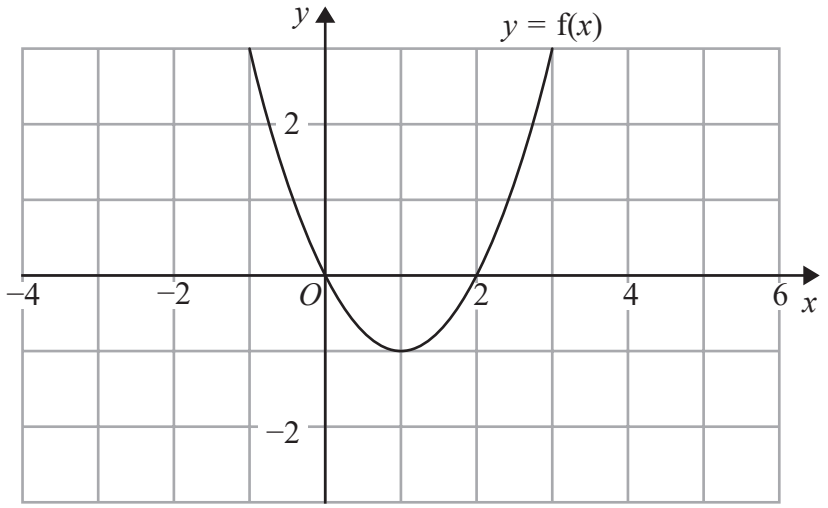
(2)

(Total for Question 14 is 4 marks)

16 Prove algebraically that $0.2\dot{5}\dot{6}$ can be written as $\frac{127}{495}$

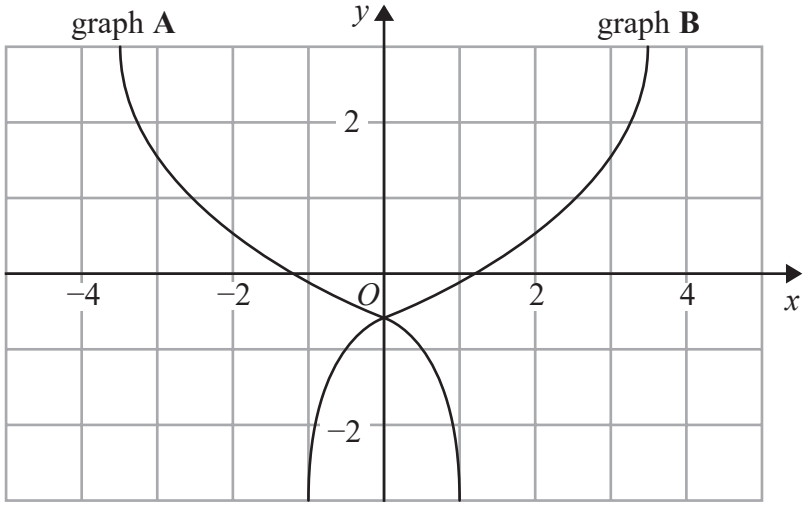
(Total for Question 16 is 3 marks)

18 The graph of $y = f(x)$ is shown on the grid below.



(*a*) On the grid above, sketch the graph of $y = f(x - 2)$

(1)



On the grid, graph **A** has been reflected to give graph **B**.

The equation of graph **A** is $y = g(x)$

(*b*) Write down the equation of graph **B**.

.....

(1)

(Total for Question 18 is 2 marks)

20 Show that $\dfrac{(\sqrt{18}+\sqrt{2})^2}{\sqrt{8}-2}$ can be written in the form $a(b+\sqrt{2})$ where a and b are integers.

(Total for Question 20 is 3 marks)

22 There are only green pens and blue pens in a box.

There are three more blue pens than green pens in the box.
There are more than 12 pens in the box.

Simon is going to take at random two pens from the box.

The probability that Simon will take two pens of the same colour is $\frac{27}{55}$

Work out the number of green pens in the box.

.....

(Total for Question 22 is 6 marks)

TOTAL FOR PAPER IS 37 MARKS